NETWORKING HOME LAB PORTFOLIO

Hands-on configuration • Verification • Troubleshooting

Project 1	MikroTik Routing and VLAN Lab
Project 2	Ubuntu Server and WireGuard VPN
Project 3	UniFi Wireless Network Integration
Project 4	Cisco Packet Tracer Networking Labs

Prepared as a practical learning portfolio for entry-level Network Technician, IT Support, and Junior Network Engineer opportunities.

This document records practical lab work and the troubleshooting methods used. It distinguishes confirmed observations from incomplete work.

Project 2 — Ubuntu Server and WireGuard VPN

Linux and VPN administration | Intel NUC running Ubuntu Server

1. Project Overview

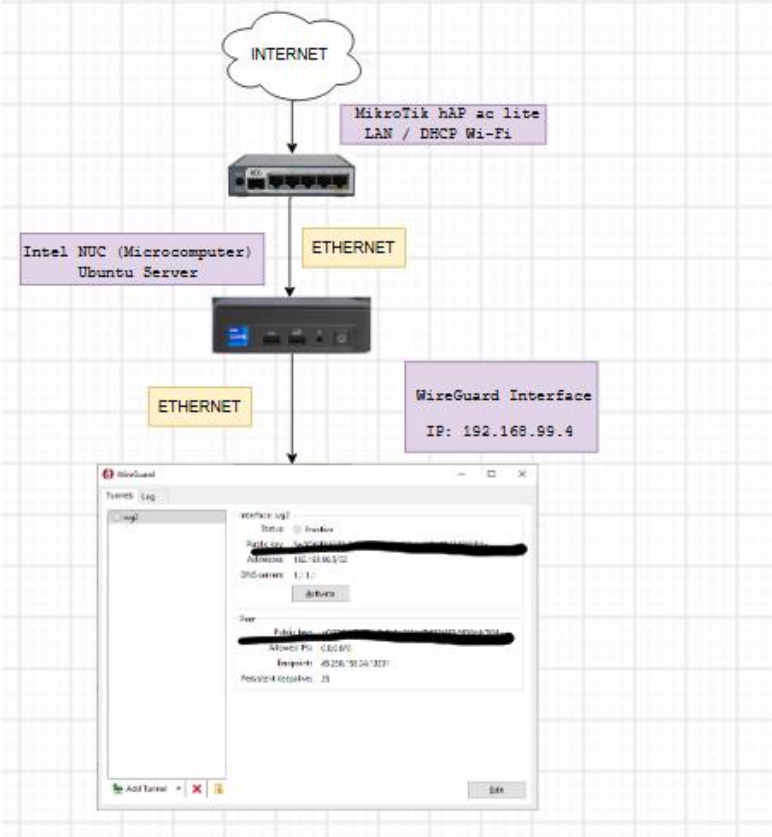
Installed and administered Ubuntu Server on an Intel NUC mini PC connected to a MikroTik hAP ac lite LAN. Practiced SSH remote administration and configured a WireGuard interface and peer settings, then investigated tunnel routing and handshake/connectivity problems.

2. Objectives

- Administer a headless Linux server from a Windows workstation using SSH.
- Inspect interfaces, IP addresses, routes, policy-routing rules, and services.
- Configure WireGuard interface and peer parameters, and manage the service with systemd.
- Troubleshoot tunnel state, handshake/transfer counters, endpoint reachability, and route selection.

3. Network Topology

Windows workstation (PowerShell/SSH) ↔ LAN/router ↔ Intel NUC running Ubuntu Server. WireGuard on the NUC is configured to connect to a remote peer endpoint. The remote peer/server is outside this project's locally controlled hardware.



4. Configuration and Commands

The following are examples/reference commands. Adapt interface names, IPs, VLAN IDs, and device versions to the configuration you actually tested.

Inspect WireGuard state

```
sudo wg show
ip addr show
ip rule
ip route show
ip route get <DESTINATION_IP>
```

Manage the systemd service (interface name used in this lab: wg)

```
sudo systemctl status wg-quick@wg
sudo systemctl restart wg-quick@wg
sudo systemctl enable wg-quick@wg
```

Example WireGuard structure — use real keys privately; never publish private keys

```
[Interface]
Address = <VPN_ADDRESS>/32
PrivateKey = <PRIVATE_KEY>

[Peer]
PublicKey = <PEER_PUBLIC_KEY>
Endpoint = <PEER_HOST_OR_IP>:<PORT>
AllowedIPs = <ROUTES_TO_SEND_THROUGH_TUNNEL>
PersistentKeepalive = 25
```

5. Verification and Testing

- Confirm the correct interface exists and check `sudo wg show` for peer and transfer information.

```
Last login: Sat Oct 10 06:31:21 2026 from 192.168.10.1
intel_nuc@ubuntuserver:~$ sudo wg show
[sudo] password for intel_nuc:
Sorry, try again.
[sudo] password for intel_nuc:
interface: wg
  public key: Ys8Y69uQy8pzJjQfnOl677/n68bNFn2z2qkRhEAaFh4=
  private key: (hidden)
  listening port: 36393
  fwmark: 0xca6c

peer: pQ9DEvnFgQVxTn5hfp/XAbrZt63Et8JTv5f0WsbTKl4=
  endpoint: 45.250.156.34:13231
  allowed ips: 0.0.0.0/0
  transfer: 0 B received, 1.45 KiB sent
  persistent keepalive: every 25 seconds
intel_nuc@ubuntuserver:~$ ip addr show wg
4: wg: <POINTOPOINT,NOARP,UP,LOWER_UP> mtu 1420 qdisc noqueue state UNKNOWN group default qlen 1000
    link/none
    inet 192.168.99.4/32 scope global wg
       valid_lft forever preferred_lft forever
intel_nuc@ubuntuserver:~$
```

- Check service state with `systemctl status wg-quick@wg`.

```
intel_nuc@ubuntuserver:~$ sudo systemctl status wg-quick@wg
    wg-quick@wg.service - WireGuard via wg-quick(8) for wg
     Loaded: loaded (8;;file://ubuntuserver/usr/lib/systemd/system/wg-quick@.service/usr/lib/systemd/system/wg-quick@.service8;;; enabled; preset: enabled)
     Active: active (exited) since Sat 2026-10-10 11:19:28 UTC; 4min 1s ago
       Docs: 8;;man:wg-quick(8)man:wg-quick(8)8;;
             8;;man:wg(8)man:wg(8)8;;
             8;;https://www.wireguard.com/https://www.wireguard.com/8;;
             8;;https://www.wireguard.com/quickstart/https://www.wireguard.com/quickstart/8;;
             8;;https://git.zx2c4.com/wireguard-tools/about/src/man/wg-quick.8https://git.zx2c4.com/wireguard-tools/about/src/man/wg-quick.88;;
             8;;https://git.zx2c4.com/wireguard-tools/about/src/man/wg.8https://git.zx2c4.com/wireguard-tools/about/src/man/wg.88;;
    Process: 862 ExecStart=/usr/bin/wg-quick up wg (code=exited, status=0/SUCCESS)
   Main PID: 862 (code=exited, status=0/SUCCESS)
        CPU: 72ms

Oct 10 11:19:27 ubuntuserver wg-quick[862]: [#] ip -4 address add 192.168.99.4/32 dev wg
Oct 10 11:19:28 ubuntuserver wg-quick[862]: [#] ip link set mtu 1420 up dev wg
Oct 10 11:19:28 ubuntuserver wg-quick[900]: [#] resolvconf -a wg -m 0 -x
Oct 10 11:19:28 ubuntuserver wg-quick[862]: [#] wg set wg fwmark 51820
Oct 10 11:19:28 ubuntuserver wg-quick[862]: [#] ip -4 rule add not fwmark 51820 table 51820
Oct 10 11:19:28 ubuntuserver wg-quick[862]: [#] ip -4 rule add table main suppress_prefixlength 0
Oct 10 11:19:28 ubuntuserver wg-quick[862]: [#] ip -4 route add 0.0.0.0/0 dev wg table 51820
Oct 10 11:19:28 ubuntuserver wg-quick[862]: [#] sysctl -q net.ipv4.conf.all.src_valid_mark=1
Oct 10 11:19:28 ubuntuserver wg-quick[862]: [#] nft -f /dev/fd/63
Oct 10 11:19:28 ubuntuserver systemd[1]: Finished wg-quick@wg.service - WireGuard via wg-quick(8) for wg.
intel_nuc@ubuntuserver:~$
```

Activate W
Go to Settings

- Use `ip rule`, `ip route`, and `ip route get <destination>` to see which route will be used.

```
intel_nuc@ubuntuserver:~$ ip -br addr
lo               UNKNOWN        127.0.0.1/8 ::1/128
eno1             UP             192.168.10.200/24 metric 100 fe80::96c6:91ff:fe11:db7a/64
wlp58s0          DOWN
wg               UNKNOWN        192.168.99.4/32
intel_nuc@ubuntuserver:~$ ip rule
0:      from all lookup local
32764:  from all lookup main suppress_prefixlength 0
32765:  not from all fwmark 0xca6c lookup 51820
32766:  from all lookup main
32767:  from all lookup default
intel_nuc@ubuntuserver:~$ ip route
default via 192.168.10.1 dev eno1 proto dhcp src 192.168.10.200 metric 100
192.168.10.0/24 dev eno1 proto kernel scope link src 192.168.10.200 metric 100
192.168.10.1 dev eno1 proto dhcp scope link src 192.168.10.200 metric 100
intel_nuc@ubuntuserver:~$ ip route show table 51820
default dev wg scope link
intel_nuc@ubuntuserver:~$
```

```
intel_nuc@ubuntuserver:~$ ip route get 192.168.20.1
192.168.20.1 dev wg table 51820 src 192.168.99.4 uid 1000
    cache
intel_nuc@ubuntuserver:~$ ip route get 8.8.8.8
8.8.8.8 dev wg table 51820 src 192.168.99.4 uid 1000
    cache
intel_nuc@ubuntuserver:~$
```

- Test the VPN peer and target network only after confirming the peer's expected routes and firewall rules.
- Compare transfer counters before and after a test; a listening interface alone does not prove a working tunnel.

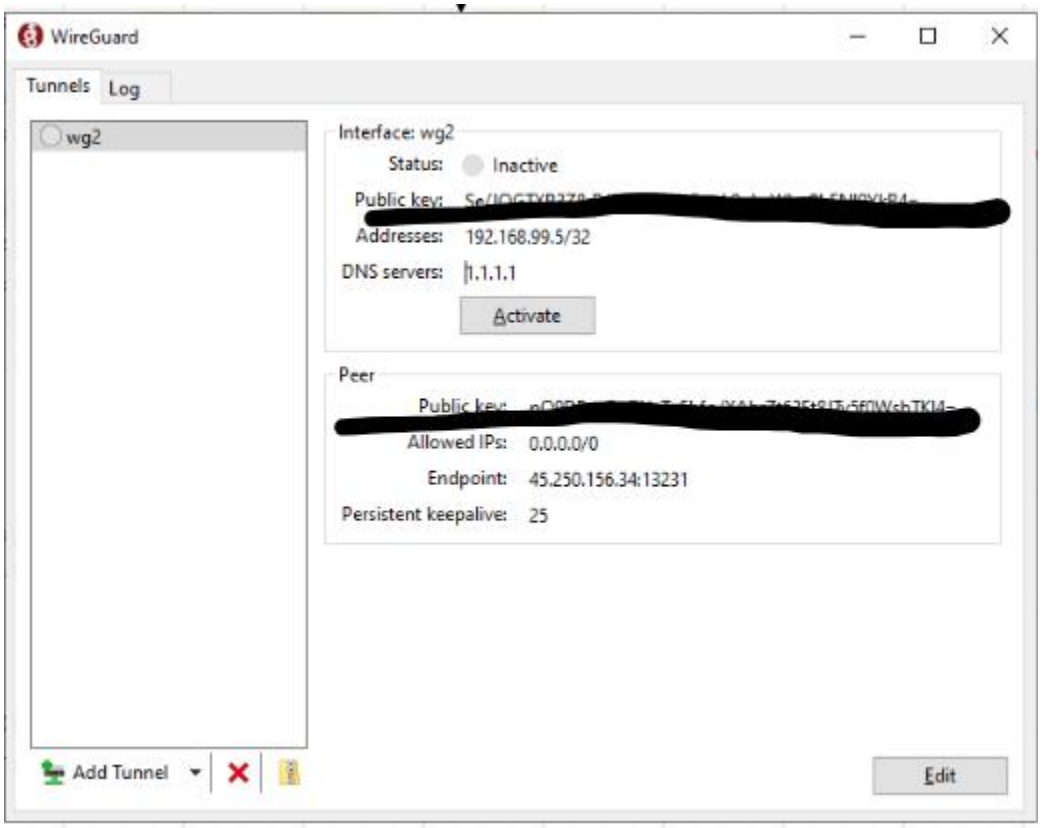
6. Troubleshooting Experience

- Investigated a WireGuard state where outbound transfer was visible but received traffic remained at 0 B, and the remote gateway could not be pinged. End-to-end connectivity was not yet verified.

```
intel_nuc@ubuntuserver:~$ sudo journalctl -u wg-quick@wg -n 50 --no-pager
Oct 10 00:27:24 ubuntuserver wg-quick[886]: [#] nft -f /dev/fd/63
Oct 10 00:27:24 ubuntuserver systemd[1]: Finished wg-quick@wg.service - WireGuard via wg-quick(8) for wg.
Oct 10 03:45:07 ubuntuserver systemd[1]: Stopping wg-quick@wg.service - WireGuard via wg-quick(8) for wg...
Oct 10 03:45:07 ubuntuserver wg-quick[1749]: [#] ip -4 rule delete table 51820
Oct 10 03:45:07 ubuntuserver wg-quick[1749]: [#] ip -4 rule delete table main suppress_prefixlength 0
Oct 10 03:45:07 ubuntuserver wg-quick[1749]: [#] ip link delete dev wg
Oct 10 03:45:07 ubuntuserver wg-quick[1749]: [#] resolvconf -d wg -f
Oct 10 03:45:07 ubuntuserver wg-quick[1749]: [#] nft -f /dev/fd/63
Oct 10 03:45:07 ubuntuserver systemd[1]: wg-quick@wg.service: Deactivated successfully.
Oct 10 03:45:07 ubuntuserver systemd[1]: Stopped wg-quick@wg.service - WireGuard via wg-quick(8) for wg.
-- Boot 62b48f5d07b047b5bdd977fc5ccb14bc --
Oct 10 03:47:57 ubuntuserver systemd[1]: Starting wg-quick@wg.service - WireGuard via wg-quick(8) for wg...
Oct 10 03:47:57 ubuntuserver wg-quick[859]: [#] ip link add wg type wireguard
Oct 10 03:47:57 ubuntuserver wg-quick[859]: [#] wg setconf wg /dev/fd/63
Oct 10 03:47:57 ubuntuserver wg-quick[859]: [#] ip -4 address add 192.168.99.4/32 dev wg
Oct 10 03:47:57 ubuntuserver wg-quick[891]: RTNETLINK answers: Network is unreachable
Oct 10 03:47:57 ubuntuserver wg-quick[859]: [#] ip link set mtu 1420 up dev wg
Oct 10 03:47:57 ubuntuserver wg-quick[896]: [#] resolvconf -a wg -m 0 -x
Oct 10 03:47:57 ubuntuserver wg-quick[859]: [#] wg set wg fwmark 51820
Oct 10 03:47:57 ubuntuserver wg-quick[859]: [#] ip -4 rule add not fwmark 51820 table 51820
Oct 10 03:47:57 ubuntuserver wg-quick[859]: [#] ip -4 rule add table main suppress_prefixlength 0
Oct 10 03:47:57 ubuntuserver wg-quick[859]: [#] ip -4 route add 0.0.0.0/0 dev wg table 51820
Oct 10 03:47:57 ubuntuserver wg-quick[859]: [#] sysctl -q net.ipv4.conf.all.src_valid_mark=1
Oct 10 03:47:57 ubuntuserver wg-quick[859]: [#] nft -f /dev/fd/63
Oct 10 03:47:57 ubuntuserver systemd[1]: Finished wg-quick@wg.service - WireGuard via wg-quick(8) for wg.
-- Boot 899916af484f46e487778f390d8efda7 --
Oct 10 08:50:40 ubuntuserver systemd[1]: Starting wg-quick@wg.service - WireGuard via wg-quick(8) for wg...
Oct 10 08:50:41 ubuntuserver wg-quick[864]: [#] ip link add wg type wireguard
Oct 10 08:50:41 ubuntuserver wg-quick[864]: [#] wg setconf wg /dev/fd/63
Oct 10 08:50:41 ubuntuserver wg-quick[864]: [#] ip -4 address add 192.168.99.4/32 dev wg
Oct 10 08:50:41 ubuntuserver wg-quick[864]: [#] ip link set mtu 1420 up dev wg
Oct 10 08:50:41 ubuntuserver wg-quick[899]: [#] resolvconf -a wg -m 0 -x
Oct 10 08:50:41 ubuntuserver wg-quick[864]: [#] wg set wg fwmark 51820
Oct 10 08:50:41 ubuntuserver wg-quick[864]: [#] ip -4 rule add not fwmark 51820 table 51820
Oct 10 08:50:41 ubuntuserver wg-quick[864]: [#] ip -4 rule add table main suppress_prefixlength 0
Oct 10 08:50:41 ubuntuserver wg-quick[864]: [#] ip -4 route add 0.0.0.0/0 dev wg table 51820
```

```
intel_nuc@ubuntuserver:~$ sudo ls -l /etc/wireguard/
sudo sudo cat /etc/wireguard/wg.conftotal 20
-rw------- 1 root root  45 Oct  7 09:07 privatekey
-rw-r--r-- 1 root root  45 Oct  7 09:14 publickey
-rw-r--r-- 1 root root 187 Oct  7 06:16 wg0.conf
-rw------- 1 root root 254 Oct  8 13:48 wg.conf
-rw-r--r-- 1 root root 276 Oct  8 05:42 wg.conf.save
intel_nuc@ubuntuserver:~$
```

- Inspected policy-routing rules and route lookup to understand why traffic to a private gateway was being sent through the WireGuard routing table.
- Resolved an SSH known-host warning by removing the stale host-key entry and reconnecting only after verifying the server identity.



7. Final Result / Current Status

Ubuntu Server and the WireGuard service/interface were configured for practice. At the latest recorded stage, remote gateway reachability and received WireGuard traffic were still unresolved. Present this as a configuration and troubleshooting project, not as a successfully verified end-to-end VPN deployment.

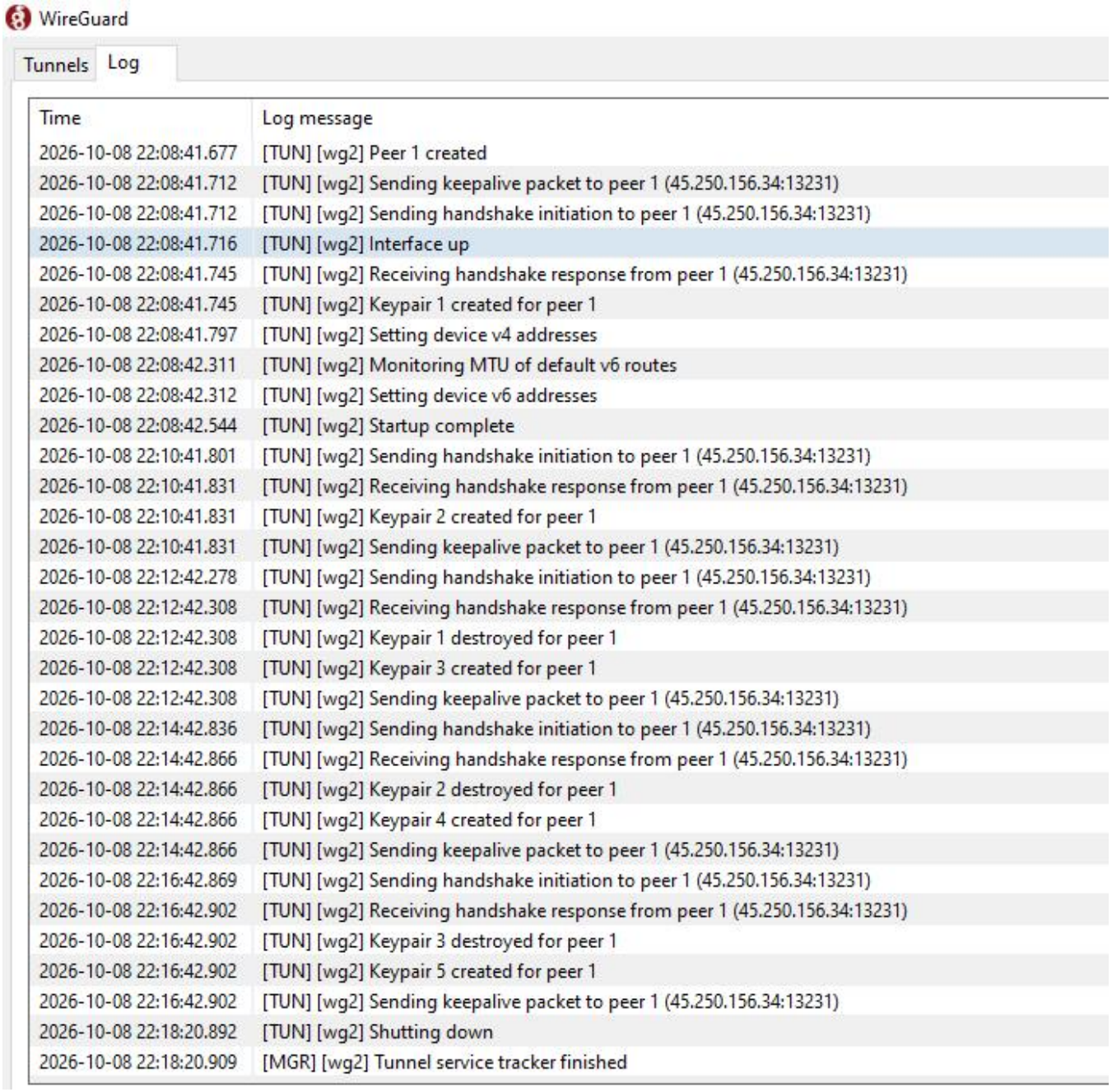
```
intel_nuc@ubuntuserver:~$ hostnamectl
 Static hostname: ubuntuserver
       Icon name: computer-desktop
         Chassis: desktop
      Machine ID: 81c8db037c594a6eabde2d95fb4f831c
         Boot ID: 294d574694f7458f8891834b73371eb2
Operating System: 8;;https://www.ubuntu.com/Ubuntu 24.04.5 LTS8;;
          Kernel: Linux 6.8.0-146-generic
    Architecture: x86-64
 Hardware Vendor: Intel Corporation
  Hardware Model: NUC7i5BNB
Firmware Version: BNKBL357.86A.0045.2017.0417.1728
   Firmware Date: Mon 2017-04-17
    Firmware Age: 9y 5month 3w 3d
intel_nuc@ubuntuserver:~$ ip -br addr
lo               UNKNOWN        127.0.0.1/8 ::1/128
eno1             UP             192.168.10.200/24 metric 100 fe80::96c6:91ff:fe11:db7a/64
wlp58s0          DOWN
wg               UNKNOWN        192.168.99.4/32
intel_nuc@ubuntuserver:~$ sudo wg show
systemctl status wg-quick@wg --no-pager
ip rule
ip route
ip route show table 51820
ping -c 4 192.168.20.1interface: wg
  public key: Ys8Y69uQy8pzJjQfnOl677/n68bNFn2z2qkRhEAaFh4=
  private key: (hidden)
  listening port: 36393
  fwmark: 0xca6c

peer: pQ9DEvnFgQVxTn5hfp/XAbrZt63Et8JTv5f0WsbTKl4=
  endpoint: 45.250.156.34:13231
  allowed ips: 0.0.0.0/0
  transfer: 0 B received, 46.97 KiB sent
  persistent keepalive: every 25 seconds
intel_nuc@ubuntuserver:~$
```

WireGuard

Tunnels | Log

Time	Log message
2026-10-08 21:59:36.656	[MGR] Starting WireGuard/1.1.1 (Windows 10.0.19045; amd64)
2026-10-08 21:59:36.676	[MGR] Starting UI process for user 'TAKLE@DESKTOP-C7HCUCL' for session 1
2026-10-08 22:08:38.586	[TUN] [wg2] Starting WireGuard/1.1.1 (Windows 10.0.19045; amd64)
2026-10-08 22:08:38.586	[TUN] [wg2] Watching network interfaces
2026-10-08 22:08:38.589	[TUN] [wg2] Resolving DNS names
2026-10-08 22:08:38.589	[TUN] [wg2] Creating network adapter
2026-10-08 22:08:39.008	[TUN] [wg2] Installing driver 1.1
2026-10-08 22:08:39.009	[TUN] [wg2] Extracting driver
2026-10-08 22:08:39.023	[TUN] [wg2] Installing driver
2026-10-08 22:08:40.251	[TUN] [wg2] Creating adapter
2026-10-08 22:08:41.647	[TUN] [wg2] Using WireGuardNT/1.1
2026-10-08 22:08:41.647	[TUN] [wg2] Enabling firewall rules
2026-10-08 22:08:41.292	[TUN] [wg2] Interface created
2026-10-08 22:08:41.675	[TUN] [wg2] Dropping privileges
2026-10-08 22:08:41.676	[TUN] [wg2] Setting interface configuration
2026-10-08 22:08:41.716	[TUN] [wg2] Monitoring MTU of default v4 routes
2026-10-08 22:08:41.677	[TUN] [wg2] Peer 1 created
2026-10-08 22:08:41.712	[TUN] [wg2] Sending keepalive packet to peer 1 (45.250.156.34:13231)
2026-10-08 22:08:41.712	[TUN] [wg2] Sending handshake initiation to peer 1 (45.250.156.34:13231)
2026-10-08 22:08:41.716	[TUN] [wg2] Interface up
2026-10-08 22:08:41.745	[TUN] [wg2] Receiving handshake response from peer 1 (45.250.156.34:13231)
2026-10-08 22:08:41.745	[TUN] [wg2] Keypair 1 created for peer 1
2026-10-08 22:08:41.797	[TUN] [wg2] Setting device v4 addresses
2026-10-08 22:08:42.311	[TUN] [wg2] Monitoring MTU of default v6 routes
2026-10-08 22:08:42.312	[TUN] [wg2] Setting device v6 addresses
2026-10-08 22:08:42.544	[TUN] [wg2] Startup complete
2026-10-08 22:10:41.801	[TUN] [wg2] Sending handshake initiation to peer 1 (45.250.156.34:13231)
2026-10-08 22:10:41.831	[TUN] [wg2] Receiving handshake response from peer 1 (45.250.156.34:13231)
2026-10-08 22:10:41.831	[TUN] [wg2] Keypair 2 created for peer 1
2026-10-08 22:10:41.831	[TUN] [wg2] Sending keepalive packet to peer 1 (45.250.156.34:13231)
2026-10-08 22:12:42.278	[TUN] [wg2] Sending handshake initiation to peer 1 (45.250.156.34:13231)



For more information about my troubleshooting experience , you may see it to my files thank you!

Time	Log message
2026-10-08 22:10:41.831	[TUN] [wg2] Sending keepalive packet to peer 1 (45.250.156.34:13231)
2026-10-08 22:12:42.278	[TUN] [wg2] Sending handshake initiation to peer 1 (45.250.156.34:13231)
2026-10-08 22:12:42.308	[TUN] [wg2] Receiving handshake response from peer 1 (45.250.156.34:13231)
2026-10-08 22:12:42.308	[TUN] [wg2] Keypair 1 destroyed for peer 1
2026-10-08 22:12:42.308	[TUN] [wg2] Keypair 3 created for peer 1
2026-10-08 22:12:42.308	[TUN] [wg2] Sending keepalive packet to peer 1 (45.250.156.34:13231)
2026-10-08 22:14:42.836	[TUN] [wg2] Sending handshake initiation to peer 1 (45.250.156.34:13231)
2026-10-08 22:14:42.866	[TUN] [wg2] Receiving handshake response from peer 1 (45.250.156.34:13231)
2026-10-08 22:14:42.866	[TUN] [wg2] Keypair 2 destroyed for peer 1
2026-10-08 22:14:42.866	[TUN] [wg2] Keypair 4 created for peer 1
2026-10-08 22:14:42.866	[TUN] [wg2] Sending keepalive packet to peer 1 (45.250.156.34:13231)
2026-10-08 22:16:42.869	[TUN] [wg2] Sending handshake initiation to peer 1 (45.250.156.34:13231)
2026-10-08 22:16:42.902	[TUN] [wg2] Receiving handshake response from peer 1 (45.250.156.34:13231)
2026-10-08 22:16:42.902	[TUN] [wg2] Keypair 3 destroyed for peer 1
2026-10-08 22:16:42.902	[TUN] [wg2] Keypair 5 created for peer 1
2026-10-08 22:16:42.902	[TUN] [wg2] Sending keepalive packet to peer 1 (45.250.156.34:13231)
2026-10-08 22:18:20.892	[TUN] [wg2] Shutting down
2026-10-08 22:18:20.909	[MGR] [wg2] Tunnel service tracker finished
2026-10-09 02:56:46.046	[MGR] Update checker: The server name or address could not be resolved
2026-10-09 03:02:35.360	[MGR] Update checker: The server name or address could not be resolved
2026-10-09 05:07:52.096	[MGR] Exited UI process for user 'TAKLE@DESKTOP-C7HCUCL' for session 1 with status 40010004
2026-10-09 05:07:53.140	[MGR] Starting UI process for user 'TAKLE@DESKTOP-C7HCUCL' for session 1
2026-10-09 05:07:54.073	[MGR] Exited UI process for user 'TAKLE@DESKTOP-C7HCUCL' for session 1 with status 40010004
2026-10-09 05:07:55.076	[MGR] Starting UI process for user 'TAKLE@DESKTOP-C7HCUCL' for session 1
2026-10-09 05:07:55.550	[MGR] Exited UI process for user 'TAKLE@DESKTOP-C7HCUCL' for session 1 with status 40010004
2026-10-09 05:07:56.566	[MGR] Starting UI process for user 'TAKLE@DESKTOP-C7HCUCL' for session 1
2026-10-09 05:07:57.185	[MGR] Exited UI process for user 'TAKLE@DESKTOP-C7HCUCL' for session 1 with status 40010004
2026-10-09 05:07:58.189	[MGR] Starting UI process for user 'TAKLE@DESKTOP-C7HCUCL' for session 1
2026-10-09 05:07:58.192	[MGR] Unable to start manager UI process for user 'TAKLE@DESKTOP-C7HCUCL' for session 1: Session has logged out
2026-10-09 05:08:39.035	[MGR] Starting at boot WireGuard/1.1.1 (Windows 10.0.19045; amd64)
2026-10-09 05:09:15.588	[MGR] Starting UI process for user 'TAKLE@DESKTOP-C7HCUCL' for session 1

8. What I Learned

- Linux interface and route diagnostics.
- How systemd manages a WireGuard quick interface.
- Why correct keys, endpoint, AllowedIPs, peer-side routes, firewall rules, and return paths all matter.
- Why a host-key warning should be checked rather than blindly ignored.

9. Real-World Application

The same diagnostic process is useful for remote-access VPNs, Linux server support, secure administration, and remote-network troubleshooting.

10. Evidence Checklist

☑ Topology diagram with labels and IP/VLAN plan

☑ Configuration screenshot or sanitized command output

☑ Verification output showing successful tests

☑ Troubleshooting evidence: symptom → investigation → fix → retest

☑ Date, device/software version, and brief caption